Algebraic and Differential Topology – Lecture Notes

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1 Introduction

Definition 1.1.
A topology on a set X is a collection τ of subsets of X such that:

(i) $\varnothing, X \in \tau$.

(ii) If $U_i \in \tau$ for all $i \in I$, then $\bigcup_{i\in I} U_i \in \tau$.

(iii) If $U_i \in \tau$ for all $i \in I$ where I is finite, then $\bigcap_{i\in I} U_i \in \tau$.

We call the pair (X, τ) a topological space. The elements of τ are called open sets.

Definition 1.2.
A **basis** for a topology on X is a collection $\mathcal{B}$ of subsets of X such that:

(i) $\forall x \in X$, $\exists B \in \mathcal{B}$ containing x.

(ii) If $x \in B_1 \cap B_2$ for some $B_1, B_2 \in \mathcal{B}$, then there is a basis element $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

$\mathcal{B}$ generates a topology τ on X as follows:
$U \in \tau$ if $\forall x \in U$, $\exists B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Remark 1.3.
It is important that $B \in \mathcal{B}$ is a subset of U in the last statement. If we only required $x \in B \cap U$, we would run into problems. For example consider $X = \mathbb{R}$ and $U = \{0, 1\}$. We know that $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}, a < b\}$ is a basis for the standard topology on $\mathbb{R}$. Then for $x = 0$, we can choose $B = (-1, 1)$ and for $x = 1$, we can choose $B = (0, 2)$. This would mean that U is open, which should not be the case.

Example 1.4.

- (M, d) is a metric space. Then $\mathcal{B} = \{B_r(x) : x \in M, r > 0\}$ is a basis for the metric topology on M, where $B_r(x) = \{y \in M : d(x, y) < r\}$ is the open ball of radius r around x.
- X, Y topological spaces. The product topology on $X \times Y$ is generated by the basis $\mathcal{B} = \{U \times V : U \subseteq X, V \subseteq Y \text{ open}\}$.
- subspace topology: Let (X, τ) be a topological space, $S \subseteq X$. Then the subspace topology of S is $\tau_S = \{U \cap S : U \in \tau\}$.

Definition 1.5.
U is a neighbourhood of x if $\exists V$ open such that $x \in V \subseteq U$.

Definition 1.6.
Let X, Y be topological spaces.
A map $f : X \to Y$ is continuous if $\forall U \subseteq Y$ open, $f^{-1}(U)$ is open in X.
If f is a bijection and both f and f^{-1} are continuous, then f is called a homeomorphism. In this case, we say that X and Y are homeomorphic or topologically equivalent ($X \simeq Y$).

Example 1.7.

- $f : [0,1] \to \mathbb{S}^1$ given by $f(t) = (cos(2\pi t), sin(2\pi t))$.
  f is continuous and bijective but f^{-1} is not continuous.
  $f([0,1))$ is not open in $\mathbb{S}^1$.
- $\mathbb{R} \not\simeq \mathbb{R}^2$: Assume they are homeomorphic. Then there exists a homeomorphism $f : \mathbb{R} \to \mathbb{R}^2$ with $f(0) = (a,b)$. But $f|_{\mathbb{R}\setminus\{0\}} : \mathbb{R} \setminus \{0\} \to \mathbb{R}^2 \setminus \{(a,b)\}$ is a homeomorphism. But $\mathbb{R} \setminus \{0\}$ is not connected, while $\mathbb{R}^2 \setminus \{(a,b)\}$ is connected. This is a contradiction.
- $\mathbb{R}^2 \not\simeq \mathbb{R}^3$: Assume they are diffeomorphic. Then there exists a diffeomorphism $f : \mathbb{R}^2 \to \mathbb{R}^3$. Then $f \circ f^{-1} = id_{\mathbb{R}^3}, f^{-1} \circ f = id_{\mathbb{R}^2}$ so $Df_x Df_y^{-1} = I_3$ and $Df_y^{-1} Df_x = I_2$.

2 Homotopy

Definition 2.1.
Let X, Y be topological spaces (as always in this chapter) and $f, g : X \to Y$ be continuous maps. Then f is called homotopic to g if there exists a $F : X \times I \to Y$ continuous s.t. $F(x,0) = f(x)$ and $F(x,1) = g(x)$ for all $x \in X$. We write $f \simeq_F g$ and F is called a homotopy between f and g. If $f \simeq c$ where c is a constant map, then f is called null-homotopic.

Definition 2.2.
A continuous map $f : I \to X$ is called a path in X.
Two paths $f, g : I \to X$ are called **path homotopic** if they have the same start and end points, i.e. $f(0) = g(0)$ and $f(1) = g(1)$, and there exists a continuous map $F : I \times I \to X$ s.t. $F(x,0) = f(x)$, $F(x,1) = g(x)$ for all $x \in I$, and $F(0,t) = f(0) = g(0)$, $F(1,t) = f(1) = g(1)$ for all $t \in I$. We write $f \simeq_F g$.

Remark 2.3.
This is an equivalence relation:

(i) $f \simeq_F f$ with $F(x,t) = f(x)$ for all $x, t \in I$.

(ii) $f \simeq_F g \implies g \simeq_G f$ with $G(x,t) := F(x, 1-t)$.

(iii) $f \simeq_F g$, $g \simeq_G h \implies f \simeq_H h$ with $\begin{cases} H(x,t) = F(x,2t) & 0 \le t \le \frac{1}{2} \\ H(x,t) = G(x,2t-1) & \frac{1}{2} \le t \le 1 \end{cases}$

If f is a path then we call $[f]$ the path homotopy equivalence class of f.

Example 2.4.

- Any two continuous maps $f, g : X \to \mathbb{R}^2$ are homotopic with $F(x,t) = (1-t)f(x) + tg(x)$.
- Let $X = \mathbb{R}^2 \setminus \{(0,0)\}$ and let $f, g : I \to X$ be defined by $f(t) = e^{\pi i t}$ and $g(t) = e^{-\pi i t}$. Then f and g are not path homotopic in X. Proving this is not trivial. First, we state an equivalent problem which is slightly easier to prove. Instead of two paths, we try to show that the unit circle as a path $h(t) = e^{2\pi i t}$ is not path homotopic to a constant path (w.l.o.g. $(1,0)$).

Assume there is a path homotopy $F : I \times I \to X$ such that $F(x,0) = h(x)$ and $F(x,1) = (1,0)$. W.l.o.g. we may assume $F : I \times I \to \mathbb{S}^1$: if F' is a path homotopy, then $F := \frac{x}{|x|} \circ F'$ is one too, because $\frac{x}{|x|}$ is continuous on X.

From here on we use the covering map (which we will define later (3.2)) $p : \mathbb{R} \to \mathbb{S}^1$, $p(t) = e^{2\pi i t}$. By a lemma which we will prove later (3.10), F has a lift $\tilde{F}$, i.e. a continuous map $\tilde{F} : I \times I \to \mathbb{R}$ with $p \circ \tilde{F} = F$. We can normalize $\tilde{F}$ such that $\tilde{F}(0,0) = 0$.

Because $F(0,t) = F(1,t) = F(x,1) = (1,0)$ are all mapped to 1 (we use $\mathbb{R}^2 \cong \mathbb{C}$ for convenience) and $p^{-1}(1) = \mathbb{Z}$, the maps $\tilde{F}(0,t)$, $\tilde{F}(1,t)$ and $\tilde{F}(x,1)$ are continuous with values in $\mathbb{Z}$, and $\mathbb{Z}$ is discrete. Hence each of them is constant. Since consecutive edges share a corner, the three constants agree, and we get

$$\tilde{F}(0,0) = \tilde{F}(0,1) = \tilde{F}(1,1) = \tilde{F}(1,0) = 0.$$

We have now determined every edge except the bottom one. On the bottom edge, both $x \mapsto \tilde{F}(x,0)$ and $x \mapsto x$ are continuous lifts of h. We see that

$$p\big(\tilde{F}(x,0) - x\big) = \frac{h(x)}{h(x)} = 1,$$

so the difference lies in $\mathbb{Z}$. Since the difference is continuous on the connected set I, it is constant. At $x = 0$ the difference is 0, hence $\tilde{F}(x,0) = x$. But this is a contradiction, because $\tilde{F}(1,0) = 0 \neq 1$.

Definition 2.5.
Let f be a path in X from x_0 to x_1 and g be a path in X from x_1 to x_2. Then the concatenation of f and g is the path $f * g : I \to X$ defined by

$$f * g(x) = \begin{cases} f(2x) & 0 \le x \le \frac{1}{2} \\ g(2x-1) & \frac{1}{2} \le x \le 1 \end{cases}$$

This induces a product on the path homotopy equivalence classes: $[f]*[g] := [f*g]$.

Theorem 2.6.

(i) $[f] * ([g] * [h]) = ([f] * [g]) * [h]$

(ii) *Let $x \in X$ and e_x be the constant path at x. Let f be a path from x to y. Then $[e_x] * [f] = [f]$ and $[f] * [e_y] = [f]$.*

(iii) Let f be a path from x_0 to y. Then there exists an inverse path $\bar{f}$ from y to x_0 defined by $\bar{f}(x) = f(1-x)$. Then $[f] * [\bar{f}] = [e_{x_0}]$ and $[\bar{f}] * [f] = [e_y]$.

Proof.

(i) $[f] * ([g] * [h]) = [f * (g * h)] = [(f * g) * h] = ([f] * [g]) * [h]$ where the second equality follows from a reparametrization of the homotopy.

(ii) Let e_0 be the constant path at I and $id = i$ on I. We see that $i \simeq e_0 * i$. Then $f = f \circ i \simeq f \circ (e_0 * i) = (f \circ e_0) * (f \circ i) = e_{x_0} * f$. A similar argument shows $[f] * [e_y] = [f]$.

(iii) Let f be a path from x_0 to y. Then $\bar{f}(x) = f(1-x)$ is a path from y to x_0. It follows that $(f * \bar{f}(x)) = \begin{cases} f(2x) & 0 \le x \le \frac{1}{2} \\ f(2-2x) & \frac{1}{2} \le x \le 1 \end{cases} = f(t|1-2|x-\frac{1}{2}||)$. Let $F : I \times I \to X$ be defined by $F(x,t) = f(t|1-2|x-\frac{1}{2}||)$. Then $F(x,0) = f(0)$ and $F(x,1) = f(|1-2|x-\frac{1}{2}||)$. It follows that $(f * \bar{f}) \simeq_F e_{x_0}$. A similar argument shows $[\bar{f}] * [f] = [e_y]$.

□

Definition 2.7.
A **loop** based at $x_0 \in X$ is a path $f : I \to X$ with $f(0) = f(1) = x_0$.
$\pi_1(X, x_0)$ is the so called **fundamental group of X relative to** $x_0 (= \{[f] : f$ is a loop based at $x_0\})$. This is a group.

Example 2.8.
$\pi_1(\mathbb{R}^n, x_0) = 0$ for all $x_0 \in \mathbb{R}^n$. Also, $\pi_1(X, x_0) = 0$ for all $x_0 \in X$ if $X \subseteq \mathbb{R}^n$ is convex.

Question: Does $\pi_1(X, x_0)$ depend on the choice of x_0?

Definition 2.9.
Let f be a loop based at x_0. We define $\alpha : [0,1] \to X$ as a path from x_0 to x_1. It is easy to see that $\bar{\alpha} * f * \alpha$ is a loop based at x_1.
We define a map $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ by $\hat{\alpha}([f]) = [\bar{\alpha} * f * \alpha]$.

Theorem 2.10.
$\hat{\alpha}$ is a group isomorphism.

Proof.
Let f, g be loops based at x_0. Then $\hat{\alpha}([f] * [g]) = \hat{\alpha}([f * g]) = [\bar{\alpha} * (f * g) * \alpha] = [\bar{\alpha} * f * e_{x_1} * g * \alpha] = [\bar{\alpha} * f * (\alpha * \bar{\alpha}) * g * \alpha] = [\bar{\alpha} * f * \alpha] * [\bar{\alpha} * g * \alpha] = \hat{\alpha}([f]) * \hat{\alpha}([g])$.
If $\beta = \bar{\alpha}$, then $\hat{\beta} = \hat{\alpha}^{-1}$. Because $\hat{\alpha}$ is a homomorphism and it has an inverse, it is a group isomorphism. □

Corollary 2.11.
Let X be path connected. Then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$ for all $x_0, x_1 \in X$.

Definition 2.12.
X is called simply connected if it is path connected and $\pi_1(X, x_0) = 0$ for some (and hence all) $x_0 \in X$.

Theorem 2.13.
Let $h : (X, x_0) \to (Y, y_0)$ be a continuous map with $h(x_0) = y_0$. Then h induces a homomorphism $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ with $h_*([f]) = [h \circ f]$ for all $[f] \in \pi_1(X, x_0)$.

Proof.
h_* is well defined: if $f \simeq g$ then $h \circ f \simeq h \circ g$.
h_* is a homomorphism: let f, g be loops based at x_0. Then $h_*([f] * [g]) = h_*([f * g]) = [h \circ (f * g)] = [h \circ f * h \circ g] = [h \circ f] * [h \circ g] = h_*([f]) * h_*([g])$. □

Theorem 2.14.
If h is a homeomorphism then h_* is an isomorphism.

Proof.
Let $k = h^{-1}$. Then $k_* \circ h_* = (k \circ h)_* = id_* = id$. □

3 Covering Spaces

Definition 3.1.
Let $p : E \to B$ be a continuous, surjective map and let $U \subseteq B$ be an open set. We say U is **evenly covered by p** if $p^{-1}(U)$ is the disjoint union of open sets V_α in E such that $p|_{V_\alpha}$ is a homeomorphism from V_α onto U for all α.
We say $\{V_\alpha\}_{\alpha \in A}$ is a partition of p^{-1} into slices.

Definition 3.2.
Let $p : E \to B$ be a continuous, surjective map. If every $b \in B$ has a neighbourhood U that is evenly covered by p, then p is called a **covering map**, B is called the **base of the covering** and E is the **covering Space of** B.
We call $p^{-1}(x)$ a fiber over x.

Remark 3.3.

- $\forall b \in B, p^{-1}(b)$ is discrete in E: each slice V_α is open in E and intersects $p^{-1}(b)$ in a single point.
- Covering maps are open: Let $V \subseteq E$ be open. Take $x \in p(V)$. Choose a neighbourhood U of x in B which is evenly covered by p. Let $\{V_\alpha\}$ be the slices of $p^{-1}(U)$ and choose $y \in V$ such that $p(y) = x$ and β such that $y \in V_\beta$. We know that $V \cap V_\beta$ is open in E, hence open in V_β. Now we know that $p|_{V_\beta} : V_\beta \to U$ is a homeomorphism, hence $p(V \cap V_\beta)$ is open in U, hence open in B and it is still a neighbourhood of x in $p(V)$.

Example 3.4.

- Let X be a topological space and $E = X \times \{1, ..., n\}$. Then $p : E \to X$ given by $p(x, i) = x$ is a covering map.
- $p : \mathbb{R} \to \mathbb{S}^1$ given by $p(x) = (\cos(2\pi x), \sin(2\pi x))$ is a covering map. For example $p^{-1}(U) = \{x \in \mathbb{R} : \cos(2\pi x) > 0\} = \bigcup_{n \in \mathbb{Z}} (n - \frac{1}{4}, n + \frac{1}{4})$, so we see that $V_n = (n - \frac{1}{4}, n + \frac{1}{4})$ are disjoint open sets. Because $\sin(2\pi x)$ is strictly monoton on $\bar{V_n}$ we get that $p|_{\bar{V_n}} : \bar{V_n} \to \bar{U}$ is bijective and also homeomorphic.

- Conversely $p : \mathbb{R}^+ \to \mathbb{S}^1$ with the same definition is **NOT** a covering map (despite being a surjective, continuous, local homeomorphism). This is due to $(1,0)$ not having a neighbourhood which is evenly covered by p so $p : V_n \to U$ is a homeomorphism for all $n \neq 0$.

Theorem 3.5.
Let $p : E \to B$ and $p' : E' \to B'$ be two covering maps. Then $p \times p' : E \times E' \to B \times B'$ is again a covering map.

Proof.
Let $b \in B, b' \in B'$ and choose U, U' to be the, by p and p' evenly covered neighbourhoods of b and b' respectively. So we get $p^{-1}(U) = \bigcup_\alpha V_\alpha$ and $p'^{-1}(U') = \bigcup_\beta V'_\beta$. It follows that $(p \times p')^{-1}(U \times U') = (\bigcup_{\alpha,\beta} V_\alpha \times V'_\beta)$ which shows that $p \times p'$ is indeed a covering map. □

Example 3.6.
- $T = \mathbb{S}^1 \times \mathbb{S}^1$ is the torus which has by the preciding theorem 3.5 a covering map in $p \times p : \mathbb{R} \times \mathbb{R} \to \mathbb{S}^1 \times \mathbb{S}^1$ given by $(s,t) \to (\cos(2\pi s), \sin(2\pi s), \cos(2\pi t), \sin(2\pi t))$.
- Let $p \times p$ be as above and let $b_0 = p(0) = (1,0)$. We also define $B_0 := (\mathbb{S}^1 \times \{b_0\}) \cup (\{b_0\} \times \mathbb{S}^1)$ and $E_0 := (p \times p)^{-1}(B_0) = (\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R})$. Then $p \times p : E_0 \to B_0$ is a covering map.

Definition 3.7.
Let $p : E \to B$ and $f : X \to B$ be a continuous map with E, B, X being topological spaces and p being a covering map. A **lifting** of f is a map $\tilde{f} : X \to E$ s.t. $p \circ \tilde{f} = f$

Example 3.8.
We choose $p : \mathbb{R} \to \mathbb{S}^1$ as before and $f : [0,1] \to \mathbb{S}^1$ given by $f(t) = (\cos(\pi t), \sin(\pi t))$. Then f lifts to $\tilde{f} : [0,1] \to \mathbb{R}$, $t \to \frac{1}{2}t$. If we look at $h : [0,1] \to \mathbb{S}^1$ given by $h(t) = (\cos(4\pi t), \sin(4\pi t))$ we see that it lifts to $\tilde{h}(s) = 2s$.

Lemma 3.9.
Let $p : E \to B$ be again a covering map, $b_0 \in B$ and $p(e_0) = b_0$. Any path $f : [0,1] \to B$ beginning at b_0 has a unique lifting to a path $\tilde{f}$ beginning at e_0.

Proof.
Let B be covered by finitley many open sets U which in turn are evenly covered by the map p (possible because $B = f([0,1])$, f is continuous, $[0,1]$ is compact and covered by the open sets $f^{-1}(U)$). Now we may choose a subdivision $0 = s_0 < s_1 < ... < s_n = 1$ of $[0,1]$ such that $f([s_i, s_{i+1}]) \subseteq U$.
Set $\tilde{f}(0) = e_0$. Suppose $\tilde{f}$ is defined for $0 \leq s \leq s_i$. We define $\tilde{f}$ on $[s_i, s_{i+1}]$ as follows:
By construction $f([s_i, s_{i+1}])$ lies in some open U which is evenly covered by p with $p^{-1}(U) = \bigcup_\alpha V_\alpha$ and $p : V_\alpha \to U$ is homeomorphic for all α. We choose β such that $\tilde{f}(s_i) \in V_\beta$ and set $\tilde{f}(s) = (p|_{V_\beta}^{-1} \circ f(s))$. Now it follows that $\tilde{f}$ is continuous and $p \circ \tilde{f} = f$ by definition.
Uniqueness: Let $\tilde{f}'$ be another lifting with $\tilde{f}'(0) = e_0$. Suppose $\tilde{f}(s) = \tilde{f}'(s) \quad \forall 0 \leq s \leq s_i$ and define V_β as before. $\tilde{f}'([s_i, s_{i+1}])$ is connected, hence it must lie entirely in V_β. Thus for $s \in [s_i, s_{i+1}]$, $\tilde{f}'(s)$ equals to some point $y \in V_\beta$ such that $p(\tilde{f}'(s)) = f(s)$ with $y \in p^{-1}(f(s))$. But there is only one such point, namely $\tilde{f}(s) \implies \tilde{f}'(s) = \tilde{f}(s)$. □

Lemma 3.10.
Let $p : E \to B$ be a covering map and $p(e_0) = b_0$. Let $F : [0,1] \times [0,1] \to B$ be a continuous map with $F(0,0) = b_0$. Then there exists a unique lifting $\tilde{F} : [0,1] \times [0,1] \to E$ such that $\tilde{F}(0,0) = e_0$. If F is a path homotopy, then so is $\tilde{F}$.

Proof.
We only provied a sketch here because this proof is very similar to the proof of the preceeding lemma 3.9. By set lemma we get that F lifts to $I \times \{0\}$ and $\{0\} \times I$. Now we impose two different subdivisions s_n and t_m as before. If F is a path homotopy, then $F(\{0\} \times I) = b_0$ and because $\tilde{F}$ is a lifting, it follows that $\tilde{F}(\{0\} \times I) \subseteq p^{-1}(b_0)$ which is discrete as we discussed. Because $\{0\} \times I$ is connected, it follows that $\tilde{F}(\{0\} \times I)$ is connected, hence $\tilde{F}(\{0\} \times I)$ is a one-point set. The same holds for $\tilde{F}(I \times \{0\})$. □

Theorem 3.11.
Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Let f, g be two paths in B from b_0 to b_1 and let $\tilde{f}, \tilde{g}$ be the liftings of f, g starting at e_0. If $f \simeq g$, then $\tilde{f}$ and $\tilde{g}$ end and start at the same point, and $\tilde{f} \simeq \tilde{g}$.

Proof.
Let $F : I \times I \to B$ be the path homotopy between f, g so $F(0,0) = b_0$. By lemma 3.10

$\tilde{F}$ is a lifting of F to E with $\tilde{F}(0,0) = e_0$ and it is also a path homotopy. We know therefore that $\tilde{F}(\{0\} \times I) = \{e_0\}$ and $\tilde{F}(\{0\} \times I) = \{e_1\}$. $\tilde{F}|_{I\times\{0\}}$ is a path starting at e_0 and the lifting of $F|_{I\times\{0\}}$. By the uniqueness of liftings $\tilde{F}(s,0) = \tilde{f}(s)$. The same argument gives $\tilde{F}(s,1) = \tilde{g}(s)$ which implies that $\tilde{f}(1) = \tilde{g}(1) = e_1$. Therefore $\tilde{f} \simeq_{\tilde{F}} \tilde{g}$. □

Definition 3.12.
Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Choose $[f] \in \pi_1(B, b_0)$. Let $\tilde{f}$ be the lifting of f such that $\tilde{f}(0) = e_0$. $\phi : \pi_1(B, b_0) \to p^{-1}(b_0) = E_{b_0}$ defined by $\phi([f]) := \tilde{f}(1)$. This ϕ is well defined due to the theorem 3.11.

Theorem 3.13.
In the same setup as in the definition 3.12: If E is path-connected then ϕ is surjective.
If E is simply connected then ϕ is bijective.

Proof.
First, let E be path-connected and $e_1 \in p^{-1}(b_0)$. $\tilde{f}$ is a path in E from e_0 to e_1. $f := p \circ \tilde{f}$ is a loop in B based at b_0. Then $\phi([f]) = \tilde{f}(1) = e_1 \implies \phi$ is surjective.
Now let E be simply connected and let $[f], [g]$ such that $\phi([f]) = \phi([g])$. Then $\tilde{f}, \tilde{g}$ are liftings that start in e_0. It follows that $\tilde{f}(1) = \tilde{g}(1)$, hence $\tilde{f} \simeq_{\tilde{F}} \tilde{g} \implies f \simeq_{p\circ\tilde{F}} g$ □

3.1 Fundamental group of circles and applications